CO-2 AT-1

1) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

a) What is the Mean of the data? what is the median

Ans

$$\text{Mean} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$= \frac{1}{27}(13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70)$$

$$= \frac{809}{27} = 29.96 \approx 30$$

$$\text{Meadian} = \frac{28}{2} = 14 \text{ value}$$

Meadian = 25

b) What is the mode of the data

Ans

25 and 35 are repeting bimodal

c) What is the midrange of the data?

Ans

$$\frac{\text{Max}-\text{Min}}{2} = \frac{70-13}{2} = \frac{83}{2} = 41.5$$

d) Can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?

Ans :-

~~$Q_1 = \frac{13+15+16+16+19+20+20+21+22+22+25+25+25}{2}$~~

$Q_1 = \frac{13+1}{2} = 7$ value

$= 20$

$Q_3 = \frac{13+1}{2} = 7$ value

$= 35$

$Q_3 - Q_1 = 35 - 20$

$= 15$

e) Give the five-number summary of the data

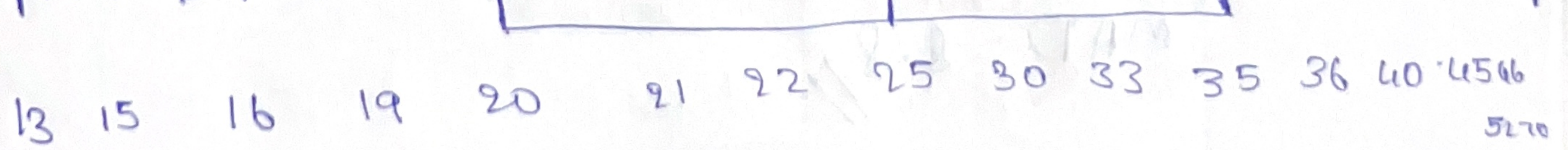
Ans

Formula : min, Q_1, median, Q_3, Max

: 13, 20, 25, 35, 70

f) Show a boxplot of the data

Ans



2) Given the following measurements for the Variable age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by the following

compute the Mean absolute deviation of age

<u>Ans</u> :-

$$\text{Mean} = \frac{18+22+25+42+28+43+33+35+56+28}{10}$$

$$= \frac{330}{10} = 33$$

(x) Age	$\lvert x-\bar{x}\rvert$
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5
	88

Mean Absolute deviation (MAD) $= \frac{\sum \lvert x-\bar{x}\rvert}{n} = \frac{88}{10} = 8.8$

mean = 33

MAD = 8.8

3) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

Ans

Total number of values = 27

~~No. of~~ bin depth = 3

Total of bin = $\frac{\text{Total}}{\text{bin depth}}$

$= \frac{27}{3} = 9$

bins = 9

Bin	Value	Bin mean	Smoothed values
B_1	13, 15, 16	14.67	14.67, 14.67, 14.67
B_2	16, 19, 20	18.33	18.33, 18.33, 18.33
B_3	20, 21, 22	21	21, 21, 21
B_4	22, 25, 25	24	24, 24, 24
B_5	25, 25, 30	26.67	26.67, 26.67, 26.67
B_6	33, 33, 35	33.67	33.67, 33.67, 33.67
B_7	35, 35, 35	35	35, 35, 35
B_8	36, 40, 45	40.33	40.33, 40.33, 40.33
B_9	46, 52, 70	56	56, 56, 56

4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

(a) min-max normalization by setting min=0 and Max=1

b) Z-score normalization.

Ans

a) Min Max Normalization (min=0, max=1)

$$x = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$= \frac{x - 200}{1000 - 200} = \frac{x - 200}{800}$$

value	x	$\bar{x}$
200	$\frac{200-200}{800}$	0
300	$\frac{300-200}{800}$	$\frac{1}{8} = 0.125$
400	$\frac{400-200}{800}$	$\frac{200}{800} = \frac{1}{4} = 0.25$
600	$\frac{600-200}{800}$	$\frac{400}{800} = \frac{1}{2} = 0.5$
1000	$\frac{1000-200}{800}$	$\frac{800}{800} = 1$

∴ 0, 0.125, 0.25, 0.5, 1

b) Z-score Normalization

$$Z = \frac{x - \mu}{\sigma}$$

Mean = 500 = μ

x	$x-\mu$	$(x-\mu)^2$
200	200 - 500 = -300	$(300)^2$
300	300 - 500 = -200	$(-200)^2$
400	400 - 500 = -100	$(-100)^2$
600	600 - 500 = 100	$(100)^2$
1000	1000 - 500 = 500	$(500)^2$

$$\sum (x-\mu)^2 = 400000$$

$$\sigma^2 = \frac{400000}{5}$$

$$\sigma = \sqrt{80000} \approx 282.84$$

x	calculation	z - score
200	200-500/282.84	-1.06
300	300-500/282.84	-0.71
400	400-500/282.84	-0.35
600	600-500/282.84	0.35
1000	1000-500/282.84	1.77

-1.06, -0.71, -0.35, 0.35
1.77